

KHOVALING, TajikistanWMO: **388460**Lat: **38.367N**Lon: **69.983E**Elev: **1468**StdP: **84.89**Time Zone: **5.00 (E05)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-9.8	-14.8	1.2	-12.3	-12.1	1.6	-8.9	7.4	-1.2	7.1	-2.3	4.7	30	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	33.9	23.2	32.5	22.3	31.3	21.6	24.5	31.7	23.5	31.0	22.5	30.1	2.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	20.7	29.6	21.2	19.0	28.8	20.1	17.7	28.4	83.8	31.5	79.1	31.4	74.9	30.2	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	6.5	6.0	-15.7	36.0	4.5	1.1	-18.9	36.8	-21.5	37.5	-24.0	38.1	-27.3	39.0		
			-16.0	25.3	4.4	1.2	-19.1	26.2	-21.7	27.0	-24.2	27.6	-27.4	28.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	-0.1	0.6	7.3	12.7	16.9	21.5	25.1	24.0	20.5	14.1	7.0	1.8
	DBStd	9.54	4.22	5.30	4.40	3.59	2.53	2.68	2.09	1.98	2.42	3.66	4.58	4.89
	HDD10.0	1070	312	266	103	17	0	0	0	0	0	11	107	256
	HDD18.3	2644	570	498	343	172	58	5	0	0	9	137	340	513
	CDD10.0	2049	0	2	19	96	214	346	468	436	313	137	17	1
	CDD18.3	582	0	0	0	1	14	102	210	177	72	5	0	0
	CDH23.3	6319	0	0	2	38	215	1101	2398	1836	642	84	3	0
	CDH26.7	2469	0	0	0	3	36	388	1105	756	169	12	0	0
Wind	WSAvg	3.4	3.5	3.4	3.4	3.3	3.3	3.3	3.3	3.4	3.6	3.5	3.4	3.4
Precipitation	PrecAvg	519	54	71	84	80	60	25	14	7	7	23	42	52
	PrecMax	736	97	121	135	187	120	54	37	21	19	50	112	100
	PrecMin	326	22	23	24	26	11	9	5	3	1	4	12	7
	PrecStd	101	21	23	27	31	29	11	7	4	4	11	25	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	19.3	21.7	26.7	29.8	33.7	36.1	34.5	30.8	28.2	22.6	16.4
		MCWB	7.4	11.0	14.1	17.9	19.4	21.7	23.6	24.1	19.6	18.0	13.6	9.2
	2%	DB	10.6	12.8	18.9	24.2	27.5	31.8	34.8	32.9	29.8	25.7	18.9	13.6
		MCWB	5.7	8.4	12.0	15.9	18.9	21.0	23.5	23.1	19.2	16.6	12.5	7.9
	5%	DB	7.9	9.7	16.4	22.1	25.6	30.5	33.3	32.0	28.7	23.3	16.3	11.1
		MCWB	4.9	6.3	11.1	15.1	18.1	21.2	22.8	22.1	18.8	15.5	10.3	6.5
	10%	DB	5.8	7.4	14.3	19.9	24.0	29.0	31.9	30.8	27.4	21.2	13.8	8.5
		MCWB	3.7	5.2	9.9	14.2	17.6	20.5	22.1	21.5	18.3	14.4	9.2	5.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	11.1	15.3	19.1	21.8	24.5	25.7	25.8	21.3	18.8	15.0	9.9
		MCDB	11.3	17.0	19.5	24.2	26.2	30.2	32.7	32.2	28.7	24.5	21.7	14.5
	2%	WB	6.6	8.5	13.0	17.6	20.0	22.9	24.5	24.6	20.5	17.6	12.4	8.3
		MCDB	8.8	11.8	17.7	21.8	24.8	29.3	32.2	31.4	28.0	24.0	17.8	12.6
	5%	WB	5.1	6.9	11.6	16.1	19.1	22.0	23.7	23.2	19.4	16.3	10.8	6.8
		MCDB	7.2	9.3	15.3	20.2	24.0	28.7	31.7	30.9	27.3	22.5	15.0	10.3
	10%	WB	3.9	5.5	10.3	14.7	18.1	21.0	22.9	22.1	18.5	14.9	9.6	5.4
		MCDB	5.9	7.2	13.4	18.9	23.0	27.4	30.9	29.9	26.5	20.3	13.3	8.1
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.4	9.5	10.7	11.8	13.0	13.0	12.5	12.4	11.7	10.4	9.5
		MCDBR	11.2	11.8	13.3	14.9	15.1	15.0	14.0	13.9	14.1	15.5	14.5	12.4
		MCWBR	7.5	7.9	8.3	8.9	8.6	8.3	7.6	8.1	8.2	9.0	8.5	7.9
	5% WB	MCDBR	9.6	10.7	12.2	13.2	13.6	14.5	13.4	13.3	12.7	13.7	13.6	11.2
		MCWBR	7.2	7.7	8.7	9.2	9.2	9.8	8.5	8.8	8.1	8.9	8.8	7.8
Clear-Sky Solar Irradiance	taub	0.331	0.377	0.439	0.454	0.466	0.488	0.483	0.443	0.403	0.420	0.384	0.332	
	taud	2.180	2.059	1.953	1.971	1.981	1.923	1.934	2.038	2.111	2.004	2.067	2.175	
	Ebn at Noon	823	826	810	825	822	802	802	826	837	766	744	788	
	Edh at Noon	106	138	170	177	179	190	186	164	143	144	119	100	
All-Sky Solar Radiation	RadAvg	2.28	3.01	4.19	5.26	6.30	7.42	7.63	7.13	5.97	3.94	2.42	1.98	
	RadStd	0.20	0.40	0.36	0.57	0.51	0.43	0.31	0.28	0.23	0.41	0.36	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A